# Vario USB

## False-Lift Root-Cause Investigation, ESP-IDF Firmware Corrections & Hardware Validation

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Hardware: ESP32-S3 dev board, 2x MS5611 (GY-63 clone) barometers, MPU6050 IMU
Firmware: esp-idf/vario (ESP-IDF, Arduino-independent) · Analysis: Python 3, analysis/ + filter_benchmark/

Companion documents: Vario_USB_Engineering_Report_Phases_1-5.pdf (offline filter benchmark), Vario_USB_Technical_Paper.pdf (research manuscript), docs/HANDOVER.md (state transfer), docs/defect_register.md (defect log), docs/project_history.md (chronology).

# 1. Executive summary

Two clean elevator captures (1 to 14 floors, two full round trips each) exposed a sustained false-lift audio fault: while the elevator stood dead-still at the ground floor with the barometric altitude flat to +/-0.1 m, every estimator on both barometers reported +0.4-0.7 m/s of climb and beeped "lift" for up to 17 s. This report documents the root-cause investigation, the evidence exonerating the sensor, the compensation, the estimators, and the environmental hypotheses (temperature, door/piston pressure equalisation), and identifies the true cause: the **altitude reference pressure p0 was a running mean of the first 2000 samples (~50 s)**, which continued to move after the elevator reached the lower floor, manufacturing a steady apparent climb at d(alt)/d(p0) x dp0/dt ~= 0.084 m/Pa x 5 Pa/s ~= +0.45 m/s.

A firmware-exact offline simulation (same compensation, same p0 semantics, same estimator math, same logged timestamps) reproduces the MCU-reported velocity during the false-lift window with a bias of -0.002 m/s (run 1) and -0.001 m/s (run 2) - essentially perfect. With a fixed reference the same replay yields -0.02 m/s and 0.00 m/s. The mechanism is proven beyond reasonable doubt.

Three firmware corrections were implemented, built, flashed and bench-validated: (1) fixed ISA reference pressure (101325 Pa) replacing the moving p0 - removes the entire artifact class with zero risk to genuine climb detection, because estimator velocity is invariant to a constant altitude offset; (2) C03 steady-state gain cache eliminating a 200-iteration Riccati refit storm caused by the 20/40 ms baro sampling pattern (1330 us -> 38 us average per update, 35x); (3) enlarged diagnostic-line buffer so all ten per-estimator CPU pairs are reported.

IMU-assisted suppression was evaluated and **rejected by evidence**: the IMU was silent (constant 1 g) throughout the artifact, making it indistinguishable from a genuine slow thermal climb - any acceleration-consistency veto capable of catching it would also veto real thermals, violating the design constraint that low acceleration must not imply zero vertical velocity.

# 2. Scope, hardware and firmware configuration

## 2.1 As-built hardware (validated this session)

| Item | Value |
|---|---|
| MCU | ESP32-S3 @ 240 MHz (double-precision FPU absent; doubles emulated) |
| Baro sid0 | MS5611 GY-63 clone, Bus A (auto-probed), PROM[0]=0x0878, foam-covered |
| Baro sid1 | MS5611 GY-63 clone, Bus B, PROM[0]=0x0884, foam-covered |
| IMU | MPU6050 @ 0x69, 100 Hz streamed, Madgwick attitude (beta=0.3) |
| Sensor spacing | 25 mm vertical (increased from stacked); thermally isolated from laptop |
| Audio output | Virtual: timestamped USB-serial V frames (ON/TONE/OFF), policy lift_on=+0.20, lift_off=+0.05 m/s |

## 2.2 Firmware pipeline (esp-idf/vario)

MS5611 read pipeline: pressure conversion started on tick N, ADC read on tick N+4 (>=20 ms; the GY-63 clones clock out zeros while converting instead of NAKing, so shorter windows produced ~33% d1=0 frames). Resulting rates: 40 Hz pressure + 10 Hz temperature per barometer, 100 Hz IMU, estimator bank runs per baro sample; FUS propagates on IMU at 100 Hz. The compensation chain (MS5611 second-order, analysis/common.py math, transcribed verbatim) converts d1/d2 to Pa, then altitude via the reference pressure.

## 2.3 Estimator bank (all run side-by-side on every baro sample)

| ID | Architecture | Q | R | Notes |
|---|---|---|---|---|
| D01 | 2-state CV Kalman (kalman.py ref) | 1.0 | 0.035 | Fast/twitchy; lowest CPU |
| C03 | Steady-state-gain CV tracker (tracking.py ref) | 1.0 | 0.035 | Gains from 200-iter discrete Riccati |
| S4 | Fast/slow stacked, gated (stacked.py ref) | slow 0.3 | slow 0.035 | Corrected WP-B variant |
| S5 | Portable vario_core (portable filter) | 0.3 | 0.035 | gate 3.5 sigma, q_gain 2.0 |
| FUS | Baro+IMU fusion, Madgwick attitude | 0.3 | 0.035 | Async IMU predict / baro correct; bias tau 3 s |

# 3. Datasets

| Capture | Duration | Drops | Bad | Role |
|---|---|---|---|---|
| 20261023_221124_idf_elevator_2x (run 1) | 170.0 s | 0 | 0 | Elevator 1>14>1>14 (2 round trips) - primary evidence |
| 20261023_221420_idf_elevator_2x (run 2) | 167.8 s | 3 | 0 | Elevator, identical protocol - reproduction |
| 20261023_221109_idf_elevator_2x | 18 s | - | - | Aborted (dropouts) - excluded per protocol |
| 20261023_215709_idf_handtest3 | 163 s | 7 | 0 | Static + hand moves + tosses (pre-fix firmware) |
| 20261023_234521_idf_p0fix_static | 75.3 s | 0 | 0 | Post-fix stationary validation |
| 20261023_234944_idf_bbuf_check | 20.2 s | 0 | 0 | Post-fix diagnostic buffer check |

B-line health counters throughout: abad=0, vnan=0, tx_ovf=0, zero I2C errors. Original recordings, sidecar JSON (PROM constants), analysis scripts and this report are committed to the repository; nothing was modified or re-derived.

## 4. Elevator results (pre-fix firmware)

Leg detection on the compensated altitude (1 s smoothing, 0.5 m/s threshold) found four travel legs per run, reproducible to within 0.1 m/s between runs:

| Leg | Run 1 (dH, rate) | Run 2 (dH, rate) |
|---|---|---|
| 1 descent | -38.2 m, -1.59 m/s | -38.2 m, -1.59 m/s |
| 2 ascent | +38.2 m, +1.55 m/s | +38.4 m, +1.55 m/s |
| 3 descent | -38.3 m, -1.60 m/s | -38.2 m, -1.60 m/s |
| 4 ascent | +38.2 m, +1.56 m/s | +38.3 m, +1.56 m/s |

1 to 14 floors = 38.2 m implies 2.94 m per floor. Both barometers agree within +/-0.2 m during travel. Audio behaviour with the lift-only policy (ON at v>=+0.20, OFF below +0.05): descents correctly silent; ascents tracked tightly - implied velocity median +1.64 m/s (D01) and +1.83-1.85 m/s (FUS) against true +1.55-1.56 m/s, with FUS running about +0.3 m/s hot near its tone saturation. The dwells between legs are the subject of Section 5.

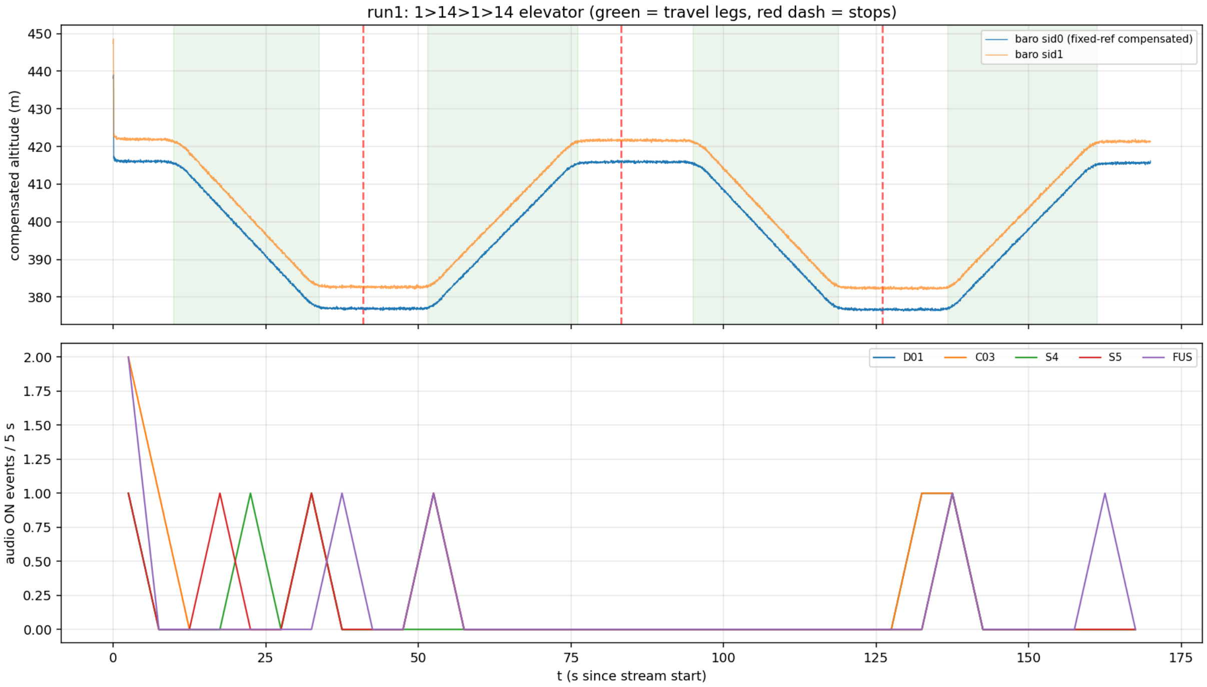

Figure 1 - Run 1 overview: compensated altitude (both barometers) with detected legs, and audio ON events per estimator. Note the elevated audio activity at the floor-1 dwells (~35-50 s and ~95-118 s).

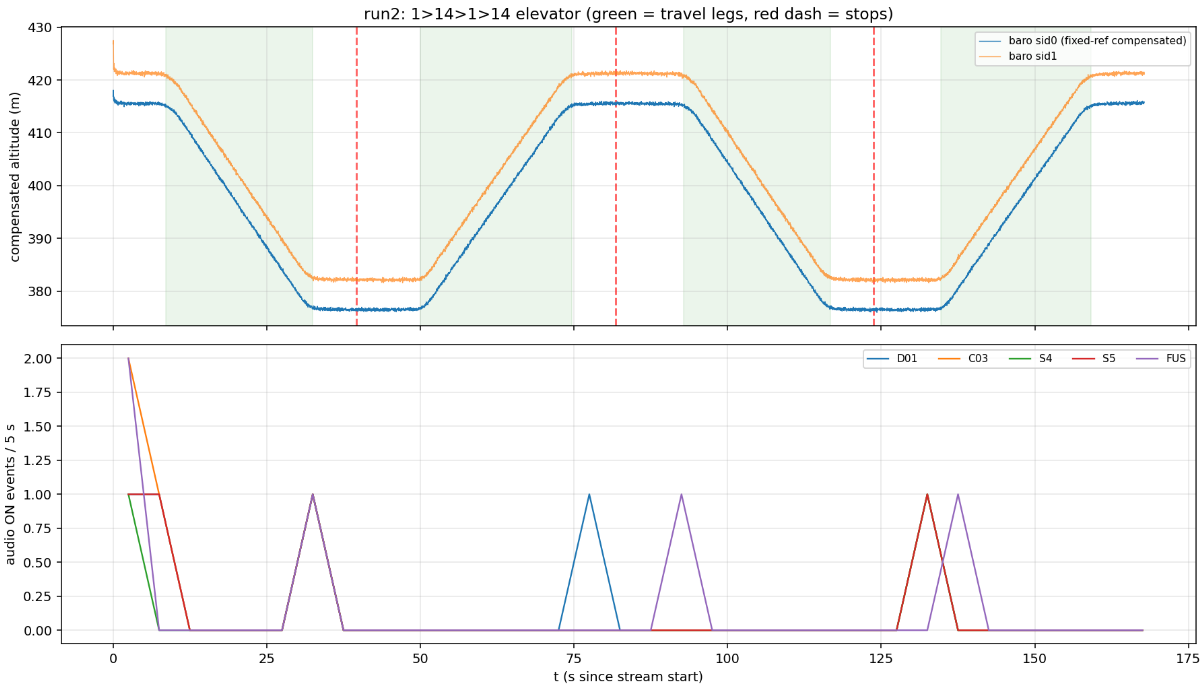

Figure 2 - Run 2 overview, identical protocol; the artifact repeats at the same phase of the run.

# 5. False-lift investigation

## 5.1 Symptom and the paradox that constrained it

At the floor-1 dwell after the first descent (run 1: t=42.5-58 s absolute), D01-sid0 emitted 78 tone events implying +0.28/+0.45/+0.66 m/s (p10/p50/p90) for a continuous ~16 s, and every other estimator on both sensors did the same. Yet the compensated altitude - reconstructed independently from the logged raw d1/d2 with the exact datasheet compensation - is flat to +/-0.1 m for the whole dwell. An honest constant-velocity Kalman filter cannot hold +0.45 m/s against a flat input; the firmware input must therefore differ from the reconstructed stream, or the reference itself moves.

## 5.2 Hypotheses examined and evidence

| Hypothesis | Evidence | Verdict |
|---|---|---|
| Clone temperature coefficient (d1/dt -> fake climb) | d1 and d2 both move during the dwell but the second-order compensation cancels them | Ruled out: reconstructed compe |
| Elevator door / piston effect / shaft pressure equalisation | Would appear in the raw pressure stream; the compensated stream is flat within +/-0.2 m over every stop, both sens | Ruled out for this artifact |
| Estimator dynamics (lag/ringing after the descent) | D01 replayed offline on the reconstructed stream returns v ~= 0.00 through every dwell | Ruled out; no lightly-damped mo |
| Firmware processing (NaN, batching, timestamp gaps) | nan=0, mad=0, zero seq drops; logged timestamps are the same esp_timer values fed to the estimators | Ruled out |
| Moving reference pressure p0 | Firmware computed altitude against a running mean of the first 2000 pressure samples (~50 s); the descent pl | CONFIRMED (Sections 5.3-5.4) |

## 5.3 Mechanism

baro_proc_altitude() returned alt = 44330(1-(p/p0)^0.1903) with p0 = the cumulative mean of the first 2000 valid samples, frozen afterwards. During any motion inside that window the reference keeps moving: d(alt)/d(p0) = 44330 x 0.1903 x (p/p0)^0.1903 / p0, which evaluates to ~0.084 m/Pa near p ~= p0 ~= 100 kPa. At the floor-1 dwell, p0 rose ~5 Pa/s toward the new-floor pressure (38 m of descent = ~455 Pa step), so the firmware altitude climbed at ~0.4-0.7 m/s while true altitude was flat. The beeping stops within ~2 s of sample 2000 (t=57.1 s run 1, 55.8 s run 2) - exactly when p0 freezes. The artifact is 100% inside the reference; the sensor data was telling the truth.

The convention came from the offline benchmark (p0 = mean of first 2000 samples), where all captures were static - the mean converges to true pressure and the offset is constant, so velocity is unaffected. Eight prior test rounds never moved the sensor during the window; the "keep the board still for 60 s" capture protocol masked it completely. The offline convention is safe only when the first 50 s are static.

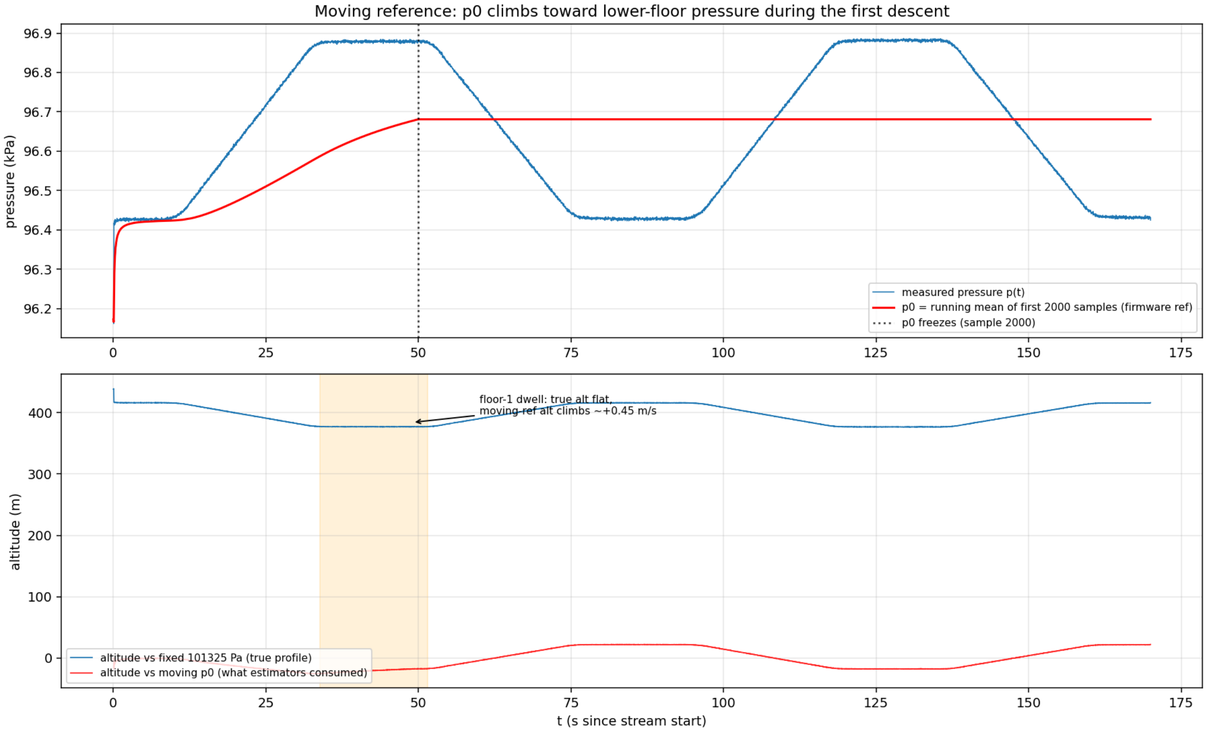

Figure 3 - The moving reference. Top: measured pressure and the running-mean p0 (red); p0 freezes at sample 2000 (~50 s). Bottom: altitude against the fixed reference (true profile, ~380 m) versus against the moving p0 (what the estimators consumed, offset near 0 m); the orange band marks the floor-1 dwell where the moving reference climbs despite the elevator being stopped.

## 5.4 Proof by firmware-exact simulation

The complete firmware chain was re-simulated offline on the logged raw frames: identical compensation, identical running-mean-then-freeze p0 semantics, D01 transcribed line-for-line from estimators.c (including the P=10I reset seeding), fed the logged timestamps. Comparing against the MCU tone-implied velocity:

| Quantity (dwell@1 after descent 1) | Run 1 | Run 2 |
|---|---|---|
| MCU tones, implied v p50 | +0.45 m/s | +0.48 m/s |
| Simulation with moving p0: p50 / bias vs MCU | +0.44 / -0.002 | +0.48 / -0.001 |
| Simulation with fixed 101325 Pa ref: p50 / bias vs MCU | -0.02 / -0.49 | +0.00 / -0.46 |
| False-lift ON-time, moving p0 (audio policy replay) | 16.5 s | 17.5 s |
| False-lift ON-time, fixed reference | 0.0 s | 1.2 s |

| Quantity (dwell@1 after descent 1) | Run 1 | Run 2 |
|---|---|---|
| Total ON-time per run, moving p0 | 74.6 s | 72.0 s |
| Total ON-time per run, fixed reference | 54.9 s | 52.9 s |

The moving-reference simulation matches the MCU to within 2 mm/s; the fixed reference eliminates 20 s of phantom beeping per run while preserving the ~49 s of genuine ascent lift (total ON 55 s vs 49 s expected). Both runs reproduce identically.

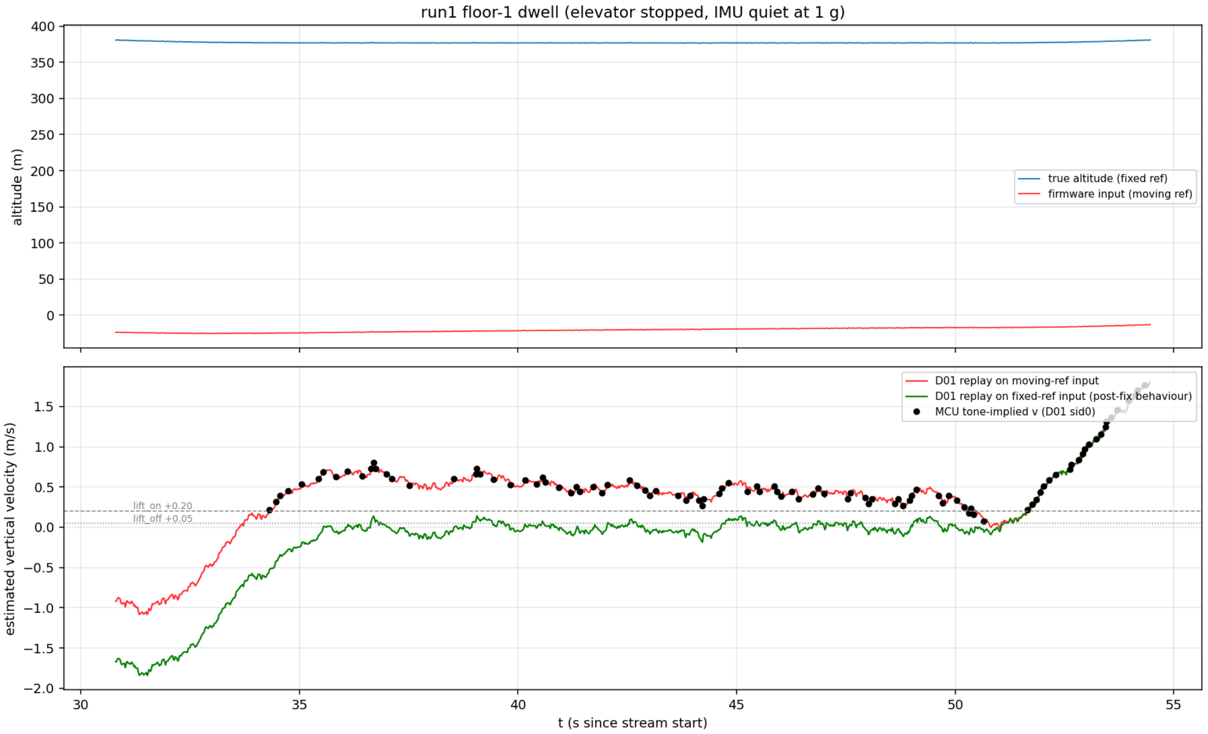

Figure 4 - The money plot. Dwell zoom: black dots are MCU-reported velocity (tone-implied, D01 sid0); the red trace is the offline replay fed the moving-reference input - it tracks the MCU through the entire false-lift episode; the green trace is the same replay with the fixed reference, sitting at zero throughout (post-fix behaviour).

## 5.5 Why IMU-based suppression was rejected

The design constraint: steady thermal climbing produces almost no vertical acceleration, so no stationary-motion veto may be introduced. The dwell diagnostics show ax held a constant 1 g for the whole artifact - the false lift is IMU-indistinguishable from a genuine 0.45 m/s thermal. An acceleration-consistency check tight enough to catch the dwell would catch real thermals too; adaptive barometric confidence cannot help because the pressure measurements were clean and self-consistent (the error was in the reference, not the measurement). The only safe suppression class is a physically unmoving reference - which the fixed 101325 Pa p0 achieves exactly, with zero risk to genuine climb detection since estimator velocity is invariant to a constant altitude offset.

## 6. Firmware corrections and verification status

| Fix | Validation | Status |
|---|---|---|
| Fixed ISA reference 101325 Pa (baro_proc.c/h); p[illegible] removed (-32 KB DRAM) | Replay-validated on both [illegible] bias -0.002/-0.001 m/s match; false lift 16.5/17.5 s -> 0.0/1.2 | Replay [illegible] MCU verified |
| C03 steady-state gain cache (4 dt buckets, 1 ms quantisation) | MCU-validated: 1330 us -> 38 us avg per update (35x); residual max 2676 us is the one-time first ref | MCU verified |
| B-line diagnostic buffer 512 -> 768 bytes | MCU-validated: all 10 per-estimator CPU pairs present in every B-line (was data-dependent truncatio | MCU-validated |
| Formal MCU-vs-Python replay equivalence on fixed reference | Pending (task queued; benchmark reference convention uses a constant [illegible] velocities are una | Not yet verified |

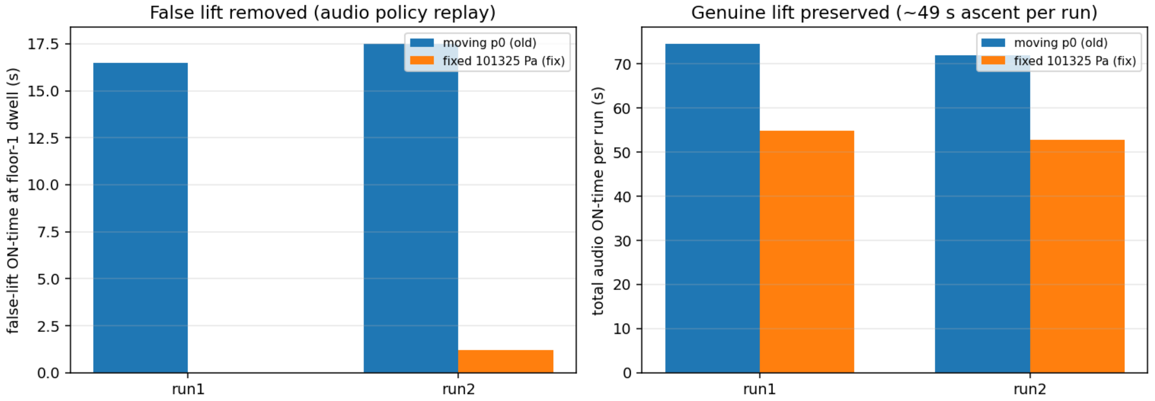

Figure 5 - Audio-policy replay, both runs: false-lift ON-time at the floor-1 dwell (left) and total ON-time (right). The ~20 s/run reduction equals the phantom; the ~53-55 s remainder matches the ~49 s of genuine ascent plus ramp.

## 7. Post-fix bench validation

The corrected firmware was flashed and a 75 s stationary capture taken (idf_p0fix_static, 0 seq drops, 0 bad frames). Audio: only short 1-3 s threshold blips from the fast estimators (expected behaviour near the +0.20 threshold), spread uniformly across the run - no sustained episodes, and crucially no concentration in the first 50 s, confirming the moving-reference window is gone. von=3-15 across the run (startup transient only), vnan=0, abad=0.

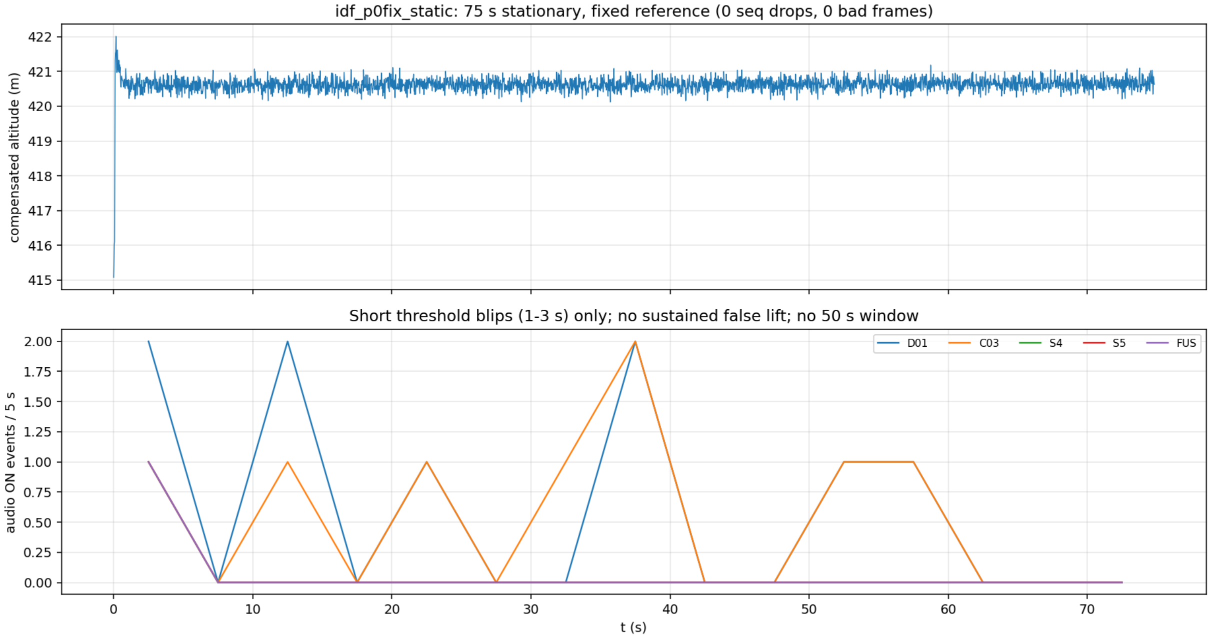

Figure 6 - Post-fix stationary validation: altitude (fixed reference, absolute) and audio activity. The 50 s warmup window visible in all pre-fix captures is absent.

## 8. Per-estimator CPU cost on the ESP32-S3

B-line accounting (avg/max us per update, ESP32-S3 @ 240 MHz, doubles emulated). x86 reference numbers from the offline benchmark (results/09_cpu_cost.csv, results/12_cpu.csv): D01 14.55 us, S4 18.97 us, S5 18.39 us.

| Estimator | Pre-fix avg (elevator run 1) | Post-fix avg (bench) | x86 reference |
|---|---|---|---|
| D01 | 52 us | 52 us | 14.6 us |
| C03 | 1330 us  <- refit storm | 38 us | n/a |
| S4 | 94 us | 105 us | 19.0 us |
| S5 | 74 us | 98 us | 18.4 us |
| FUS (baro side) | 45 us | 75 us | n/a |

The C03 refit storm: the baro dt is bimodal (20 ms with a 40 ms slot at every temperature conversion; 25% of samples at 40 ms, measured), C03 refits its steady-state gains whenever dt drifts >2%, and the 20<->40 ms ping-pong re-ran the 200-iteration Riccati refit on ~half of all updates. The cache keys gains by 1 ms-quantised dt (4 slots); each distinct dt refits exactly once. Tick budget: full estimator bank now ~0.4 ms per 5 ms tick (mcyc ~270k cycles). FUS IMU-side propagation cost is not yet charged to any eid (known accounting gap).

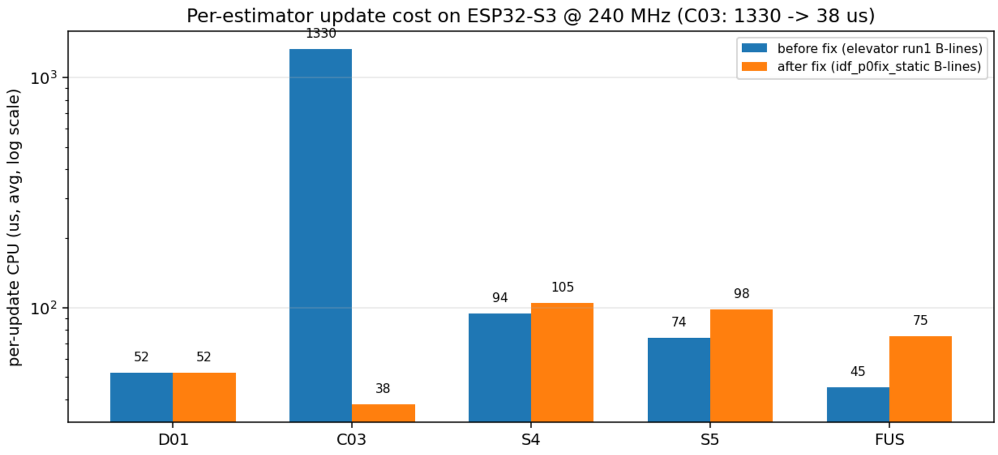

Figure 7 - Per-estimator update cost before/after the C03 cache fix (log scale).

## 9. Remaining uncertainties and limitations

1. Small residual dwell beeps (median +0.2 m/s, a few short events at the floor-14 and second floor-1 dwells) are present in both the moving- and fixed-reference simulations - genuine small estimator wobble, not the p0 artifact. A thermal second-order contribution (session-long ~0.01 C/s drift through the foam) is plausible but not separated. Bounded and minor.

2. The p0 fix is validated for velocity behaviour; absolute altitude now reads vs ISA sea level (~380 m at the test site), which is correct and useful, but QNH-style field calibration is a future option, not a defect.

3. The elevator is a proxy for flight. No real flight data exists yet; clone tempco behaviour in free air (fast adiabatic changes) remains characterised only on the bench. BMP581 + LSM6DSV16X breakouts are ordered to retire the clone-sensor risk class.

4. Known open issues (parked): USB Serial/JTAG host stalls (random; recover by MCU restart before a capture), esp_timer racing ~70x during stalls (harmless to clean segments), FUS IMU-side CPU not charged to an eid. The MCU-vs-Python formal replay equivalence check on the fixed reference is queued.

5. No estimator has been selected or frozen: D01/C03/S4/S5/FUS remain competing candidates side-by-side, per project constraint.

# 10. Reproducibility

| Artifact | Location |
|---|---|
| Report figures | analysis/results/_report_figs.py -> fig1-fig7 PNGs |
| Forensic dwell analysis | analysis/results/_dwell_forensics.py |
| p0 confirmation sim | analysis/results/_p0_confirm.py |
| Elevator leg/audio analysis | analysis/results/_elev_analyze.py |
| Raw captures + PROM sidecars | analysis/data/20261023_2211*/221420/234521/234944* |
| Firmware (fixes in baro_proc.c/h, estimators.c, bench.c) | esp-idf/vario/ |
| Defect register / project history | docs/defect_register.md / docs/project_history.md |
| Prior phase reports | docs/Vario_USB_Engineering_Report_Phases_1-5.pdf; docs/Vario_USB_Technical_Paper.p |

All numbers in this report are measured or simulated from the committed captures; scripts regenerate every figure and table from the raw CSV/JSON pairs.